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DTSA 5510 Unsupervised Algorithms in Machine Learning

Final Project Presentation

University of Colorado **Boulder**

Background and Problem Statement

- To explore the prevalent themes and narratives in news articles during the 2020 presidential election
  - aiming to understand media focus and public discourse as we approach the 2024 election

- Driven by curiosity to explore the topics and issues highlighted in news coverage during a significant political event

Problem Statement : Explore unsupervised learning techniques for analyzing thematic structure of news articles based on the text content.

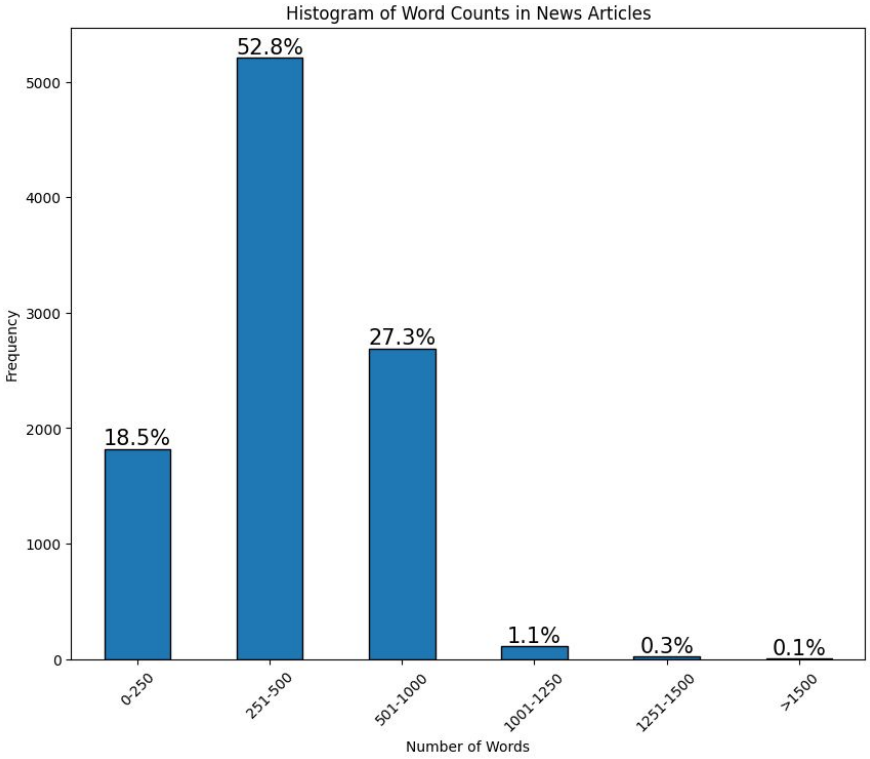
Machine Learning Approach

- Data Cleaning and Exploratory Data Analysis
- Model Selection
- Hyperparameter and Model Tuning
- Evaluation of results and learning

Data Cleaning and Exploratory Data Analysis

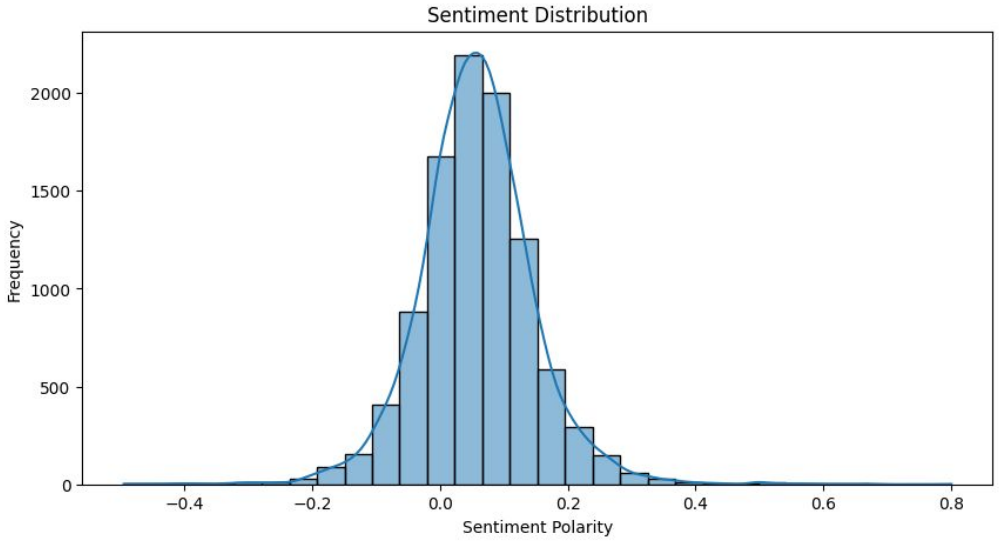
- Removed duplicates, checked for blanks
- Histogram of word counts
- Sentiment Distribution
- Named Entity Recognition

Histogram of Word Counts



Distribution suggests that most news articles are concise, with a smaller proportion of longer, more detailed articles.

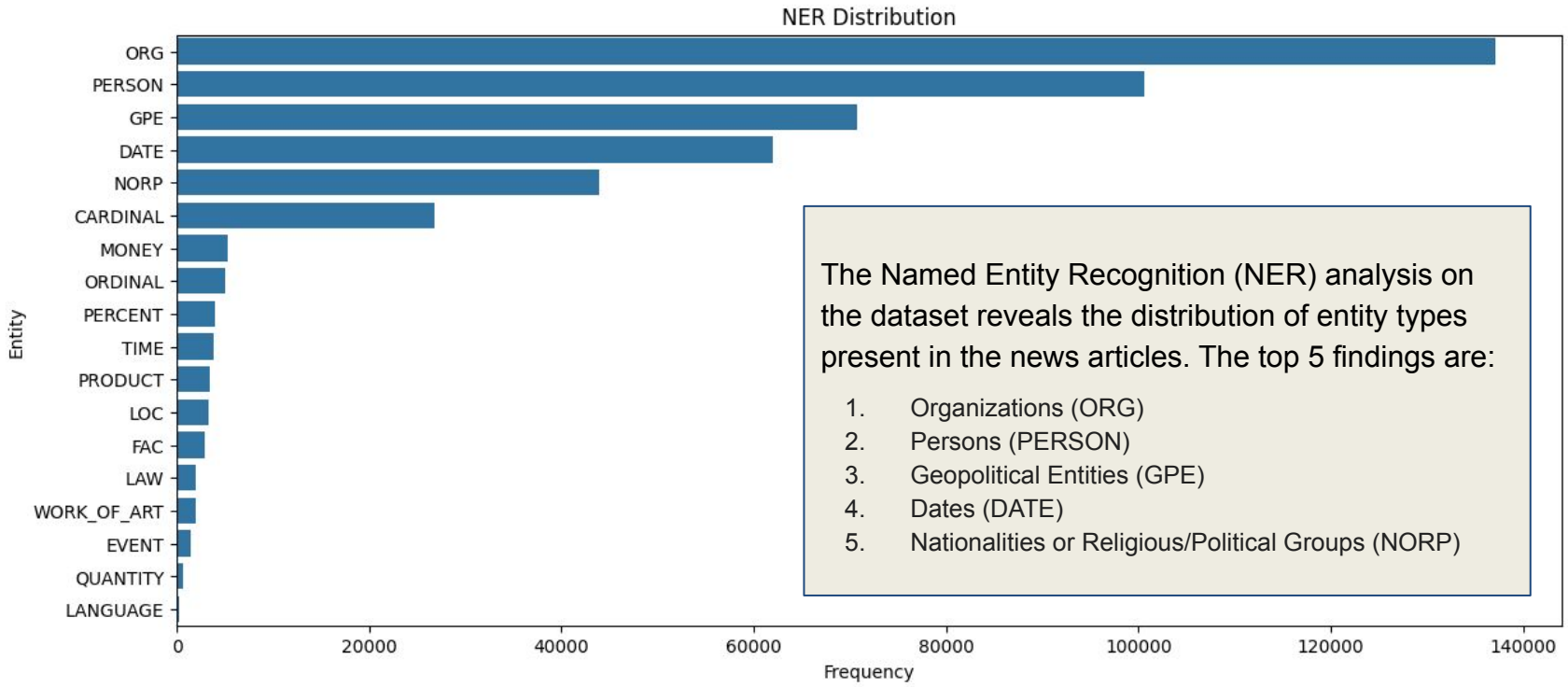
Sentiment Distribution



The histogram shows a high concentration of articles with sentiment polarity values close to zero.

This neutrality might be typical for news articles, as they often aim to present information objectively.

Named Entity Recognition



The Named Entity Recognition (NER) analysis on the dataset reveals the distribution of entity types present in the news articles. The top 5 findings are:

1. Organizations (ORG)
2. Persons (PERSON)
3. Geopolitical Entities (GPE)
4. Dates (DATE)
5. Nationalities or Religious/Political Groups (NORP)

Model Selection

Latent Dirichlet Allocation (LDA)

- Discover hidden themes or topics within the news articles.
- Preprocessing : Remove Email, Numbers, Whitespace, Single quotes, Stop words
- Number of clusters
- Coherence Score – measures the degree of semantic similarity between the top words in a topic

K-means Clustering

- Group similar news articles together based on their textual content.
- Preprocessing : Remove Email, Numbers, Whitespace, Single quotes, Stop words
- Term Frequency Thresholds to filter out very common and very rare terms
- Number of clusters
- Silhouette Score – measures the degree of separation between the clusters and the cohesion within the clusters

Evaluation of Results

Latent Dirichlet Allocation (LDA)

Topics	Silhouette Score
5	0.532
6	0.509
7	0.528
8	0.533
9	0.540
10	0.542
11	0.528
12	0.538
13	0.519
14	0.518
15	0.525

K-Means Clustering

Clusters	Coherence Score
5	0.136
6	0.105
7	-0.010
8	-0.027
9	-0.026
10	-0.035
11	-0.034
12	-0.034
13	-0.064
14	-0.034
15	-0.061

Evaluation of Results: Latent Dirichlet Allocation (LDA)

Based on number of Topics with the highest Coherence Score

- **Number of Topics: 10; Coherence Score:** 0.542
- **Topics:**
  - **Topic 1:** health, obamacare, insurance, care, healthcare, people, would, said, americans, republicans
  - **Topic 2:** people, white, said, women, one, twitter, donald, right, like, moore
  - **Topic 3:** court, said, ban, law, states, people, order, federal, immigration, administration
  - **Topic 4:** said, russia, russian, house, president, campaign, investigation, election, comey, intelligence
  - **Topic 5:** said, senate, republican, bill, house, republicans, would, democrats, congress, vote
  - **Topic 6:** said, president, united, states, north, china, korea, would, reuters, state
  - **Topic 7:** would, said, percent, tax, billion, million, new, year, budget, companies
  - **Topic 8:** election, clinton, hillary, vote, voters, republicans, people, president, campaign, percent
  - **Topic 9:** tax, said, house, would, republican, senate, bill, republicans, reform, reuters
  - **Topic 10:** donald, president, twitter, people, one, like, via, image, featured, even

Evaluation of Results: K-Means Clustering

Based on number of Topics with the highest Silhouette Score

- **Number of Clusters: 5; Coherence Score:** 0.126
- **Top Terms per Cluster:**
  - **Cluster 1:** north, korea, china, said, united, trade, states, nuclear, would, president
  - **Cluster 2:** said, would, president, house, state, republican, states, reuters, administration, senate
  - **Cluster 3:** tax, said, would, bill, house, senate, republican, republicans, percent, plan
  - **Cluster 4:** said, president, donald, people, would, one, house, white, twitter, like
  - **Cluster 5:** said, russia, russian, president, election, house, campaign, intelligence, comey, investigation

Learning

Latent Dirichlet Allocation (LDA) and K-Means Clustering are both unsupervised learning algorithms, but they are fundamentally different in terms of their approach :

	Latent Dirichlet Allocation (LDA)	**K-Means Clustering**
Approach	Identifies **hidden thematic structures** in text, where documents are mixtures of topics, and topics are mixtures of words.	Partitions data into clusters by **minimizing within-cluster variance.**
Nature of Clusters	**Probabilistic clusters**, allowing words and documents to belong to multiple topics with different probabilities.	**Hard assignments**, with each data point belonging to one distinct cluster.
Evaluation of metrics	**Coherence score**, measuring the semantic similarity of top words in a topic.	**Silhouette score**, evaluating how well-defined and distinct the clusters are.

Learning: The choice between LDA and K-Means Clustering depends on the goals and nature of the dataset

When to Use LDA:

- **Text Analysis:** Great for exploring and identifying the main topics in a collection of documents, especially when topics might overlap.
- **Interpretability:** Helps to understand the connections between different words and topics, providing a nuanced view of the content.

When to Use K-Means Clustering:

- **General Clustering:** Useful for grouping data into clear and separate categories, especially when clusters are distinct.
- **Simplicity and Speed:** Easy to set up and runs quickly, making it suitable for large datasets where fast and straightforward results are needed.